COURSE TITLE : CHEMICAL ENGINEERING
COURSE CODE : 4121
COURSE CATEGORY : B
PERIODS PER WEEK : 4
SEMESTER : 4
PERIODS PER SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	MASS TRANSFER OPERATIONS Unit operation and unit processes Distillation Extraction Test Total	14
2	Absorption and humidification Heat transfer operations Modes of heat transfer Heat transfer equipments Test Total	14
3	MECHANICAL OPERATIONS Filtration Centrifuge and mixing Size reduction Test Total	14
4	FLUID PARTICLE MECHANICS Flow of fluids Material balance Test Total	14
	Total	**56**

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand distillation operation
	2	Comprehend extraction absorption and stripping operation
2	1	Understand humidification operation
	2	Understand heat transfer operations
	3	Understand radiation and heat transfer equipments
3	1	Understand mechanical operations
	2	Understand mixing and size reduction
4	1	Understand Fluid Mechanics
	2	Comprehend particle mechanics

Specific outcome

MODULE I

1.1.0 Understand distillation operation

1.1.1 State unit operations and unit processes.
1.1.2 Explain the various chemical unit operations and unit process
1.1.3 State volatility, relative volatility, mole fraction, distillation, Roult's law Hentry's law
1.1.4 Define Distillation
1.1.5 List the types of distillation
1.1.6 Distinguish between simple and steam distillation and derive the quation State the Rayleigh's equation
1.1.7 Solve simple problems using reyleigh's equation
1.1.8 Illustrate extractive distillation and their application

1.2.0 Comprehend extraction absorption and stripping operation

1.2.1 Define extraction and state the application of extraction operation
1.2.2 Illustrate the bollman extractor and working.
1.2.3 Explain the absorption and stripping .
1.2.4 Describe the absorption towers and tower packing .

1.2.5 Explain Bubble cap, perforated and sieve tray columns .
1.2.6 Define loading point flooding point channeling over all plate efficiency point efficiency

MODULE II

2.1.0 Understand humidification operation

2.1.1 List absorption and humidification operations
2.1.2 Define various terms involved in describing humidification operation
2.1.3 State adiabatic saturation temperature
2.1.4 Explain the wet bulb temperature dry bulb temperature and explain wet bulb theory
2.1.5 Read the psychrometric chart
2.1.6 Describe the factors influencing wet bulb temperature

2.2.0 Understand heat transfer operations

2.2.1 Define Heat Transfer and Interpret various modes of heat transfer and their application
2.2.2 Define the conduction
2.2.3 Distinguish between the steady state and unsteady state heat flow
2.2.4 State the Fourier law of heat transfer by conduction
2.2.5 Define the thermal conductivity and state its unit and the factors affecting thermal conductivity
2.2.6 Explain convection, natural convection and forced convection
2.2.7 Compare the co – current and counter current methods in heat transfer process
2.2.8 Explain Infer the film concept of heat transfer temperature gradient in forced conversion
2.2.9 State the importance of re. Pr. Numbers
2.2.10 Explain nucleate boiling and film boiling

2.3.0 Understand radiation and heat transfer equipments

2.3.1 Explain the concept of radiation
2.3.2 Define absorptivity, reflectivity and emissivity.
2.3.4 Illustrate and Explain Heat Transfer Equipments
2.3.5 Describe with a neat sketch the parts of shell and tube heat exchanger (shell & tube, baffles)

MODULE III

3.1.0 Understand mechanical operations
3.1.1 Classify the filtration equipment
3.1.2 Illustrate the diagram and explain the working of sand filter, plate and frame filter and rotary continuous filter
3.1.3 List the examples of filter aid and State it's function
3.1.4 Describe centrifugal filtration with a neat sketch
3.1.5 List the application of various filters
3.1.6 Compare the constant pressure and constant rate filtration
3.1.7 Explain the working of continuous centrifuge top driven, bottom driven centrifuges
3.1.8 Describe the application of centrifuges

3.2.0 Understand mixing and size reduction

3.2.1 Illustrate the mixing equipment for liquid and liquid mixer
3.2.2 Illustrate the mixing equipment for viscous mass and stiff mass and give their working operation
3.2.3 Give the criteria for the selection mixers
3.2.4 Classify the size reduction equipment
3.2.5 State kicks law and Rittinger's law
3.2.6 Draw the line diagram and explain the working of jaw crusher, gyratory crusher hammer mill, ball mill
3.2.7 Distinguish between crushing and grinding
3.2.8 Define crushing efficiency
3.2.9 Draw the circuit diagram for closed circuit grinding

MODULE IV

4.1.0 Understand Fluid Mechanics

4.1.1 Define normality, molarity, molality weight percentage, mole percentage, and mole fraction
4.1.2 Explain flow of fluids- Study of various pipes and fittings, valves
4.1.3 Describe the classification of pipes
4.1.4 List the various types of pipe fittings
4.1.5 State the operations of various gasketing and packing materials
4.1.6 Explain the classification of valves
4.1.7 Describe the application and working globe, gate, butterfly valve check valves and safety valves
4.1.8 Classify the check valves and explain each one of them

4.2.0 Comprehend particle mechanics

4.2.1 State law of conservation of mass
4.2.2 State law of conservation of energy
4.2.3 State the material balance equation for solvent extraction
4.2.4 Define the density, specific gravity – conversion of specific gravity scales
4.2.5 Solve the problems based on above principles
4.2.6 Explain energy balance
4.2.7 Describe Law of conservation of energy
4.2.8 Define the term internal energy, enthalpy
4.2.9 Derive heat balance equation for simple process

CONTENT DETAILS

MODULE -I
MASS TRANSFER OPERATIONS

Distillation Define the distillation, mole fraction , less volatile , more volatile, relative volatility Roult's law, Hentry's Law – sketch and explain equilibrium diagram and boiling pint diagram and their importance in distillation operation . Types of distillation – simple or differential distillation, steam distillation equilibrium distillation, extractive and azeo tropic distillation, minimum number of plales by MC – cabe thiele method – simple distillation – Rayleigh equation using to solve simple problems – steam distillation and other application – elemental idea of extractive and azeotropic distillation -

Extraction Industrial importance of extraction operation list the various types solvents, qualities of solvent required – extraction equipment's – construction and working of bollman extraction and Kennedy extractor

MODULE –II

Absorption Gas and liquid velocities - define the loading point, flooding point, channeling, point efficiency overall plate efficiency and pressure drop packing towers describe various types of absorption towers types of packing materials and their qualities – packing supports, methods of packing - liquid distributors – factors controlling rate of absorption 2. Humidification Define and mathematical expression of molal humidity, absolute humidity, relative humidity, relative humidity, percentage humidity, humidity heat, humid volume, enthalpy , dry bulb temperature wet bulb temperature – psychrometry chart and its application – calculate humidity with and without psychrometry chart – wet bulb theory - application and factors influencing wet bulb termperature

MODULE -III

MECHANICAL OPERATIONS

1. Filtration. Classification of filters, atmospheric, pressure and vacuum, plate and frame – leaf filters – rotary continuous filters – construction and operation of above filters an their applications. Filter operation and effect of pressure, construct pressure and constant volume,

2. Centrifuge and mixing Classification of centrifuge and mixing equipment's – continuous centrifuge, top driven and bottom driven – operation and application types of mixing equipment – stirrers and gyitators application and their working

3. Size reduction Types of reduction equipments – blake jaw crusher, gyratory crusher, hammer mill ball mill – open and closed circuit grinding, laws of crushing – kicks and rittingers law.

MODULE IV

FLUID PARTICLE MECHANICS

Define Normality, molarity, weight percentage, mole percentage, weight fraction, mole fraction.

1. Flow of Fluids Pipes - piping stenders – classification of pipes fittings - pipe joints – gaskets and packing materials Valves: - classification of vlaves -plug (cocks) globe, gate (raising stem and non raising stem) check valves (swing, lift and ball check valves) safety valve – automatic control valve – application and working of above valves.

2. Material balance Law of conservation of mass – materials balance of equation for process with and without chemical reactions like solvent extraction mixed acid evaporation and distillation

3. Energy balance Law of conservation of energy – Heat balance equation for simple process, heat exchangers, evaporators and distillation towers

Reference

1..Unit operations of chemical engineering – M.C. Cabe de Smith, MC Mcgraw Hill
2..Introduction of chemical engineering – Badger and Banhereo
,3..Stochiometry – Bhat and Vora
4..Mass transfer operation – Robert Treybal
5..Mass & heat transfer – Raj Put,S Chand